



1. Identification of the substance/preparation and of the company undertaking

- **Trade Name:** KPL Zinc Sulphate (Heptahydrate)
 - **Application:** Zinc micronutrient fertilizer for soil and foliar application in deficient soils/crops
 - **Manufacturer/Supplier:**
Kesar Petroproducts Ltd.
Plot No. D-7/1, Lote Parshuram MIDC, Tal-Khed, Dist. Ratnagiri,
Maharashtra – 415722
 - **Further Information:** Product Safety Department, Kesar Petroproducts Ltd.,
Ratnagiri Plant
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2. Hazards Identification

- **GHS Classification:**
 - **Harmful if swallowed** (H302)
 - **Causes serious eye irritation** (H319)
 - **May cause respiratory irritation** (H335)
- **Label Elements:**
 - **Pictogram:**



(Exclamation mark)

- **Signal Word: Warning**
- **Precautionary Statements:**
 - P261 – Avoid breathing dust
 - P270 – Do not eat, drink, or smoke when using
 - P305+P351+P338 – IF IN EYES: Rinse cautiously with water for several minutes

- P337+P313 – If eye irritation persists: Get medical advice
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3. Composition/Information on Ingredients

- **Description:** Zinc Sulphate Heptahydrate
 - **Chemical Formula:** $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$
 - **CAS No.:** 7446-20-0
 - **Typical Purity (Fertilizer Grade):** $\geq 98\%$
 - **Zinc (Zn) Content:** $\geq 21\%$
 - **Sulphur (S) Content:** $\sim 11\%$
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4. First-Aid Measures

- **Inhalation:** Move to fresh air. If coughing or discomfort occurs, seek medical attention.
 - **Skin Contact:** Wash with soap and water. Remove contaminated clothing.
 - **Eye Contact:** Rinse immediately with plenty of water for at least 15 minutes. Seek medical advice.
 - **Ingestion:** Rinse mouth. **Do NOT induce vomiting.** Give water or milk if conscious. **Seek immediate medical attention.**
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5. Fire-Fighting Measures

- **Extinguishing Media:** Water spray, dry chemical, CO_2
 - **Special Hazards:** Non-combustible, but may release toxic zinc and sulphur oxides if exposed to intense fire
 - **Firefighter Protection:** Wear full protective gear and self-contained breathing apparatus
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6. Accidental Release Measures

- **Personal Precautions:** Wear gloves, goggles, and dust mask
 - **Environmental Precautions:** Prevent entry into drains, soil, or waterways (toxic to aquatic life)
 - **Cleanup:** Collect solid material mechanically. Dampen dust if necessary. Place in sealed container.
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7. Handling and Storage

- **Handling:** Avoid dust formation. Use in well-ventilated areas. Wash hands after handling.
 - **Storage:** Store in a cool, dry, well-ventilated area in tightly closed containers. Protect from moisture (hygroscopic).
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8. Exposure Controls/Personal Protection

- **Occupational Exposure Limit:**
 - **Zinc (as Zn):** ACGIH TLV-TWA 5 mg/m³ (respirable)
 - **Engineering Controls:** Local exhaust at bag opening/mixing points
 - **PPE:**
 - Respiratory: N95 mask if dust > visible levels
 - Hand Protection: Nitrile or rubber gloves
 - Eye Protection: Chemical splash goggles
 - Skin Protection: Long sleeves, apron
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9. Physical and Chemical Properties

- **Appearance:** White or colorless crystalline granules or powder
 - **Odour:** Odourless
 - **pH (5% aqueous solution):** 4.0 – 5.0
 - **Solubility in Water:** Highly soluble (~96 g/100 mL at 20°C)
 - **Specific Gravity:** ~1.97
 - **Melting Point:** Decomposes at ~70°C (loses water of crystallization)
 - **Hygroscopicity:** Yes — may cake or dissolve if exposed to humid air
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10. Stability and Reactivity

- **Stable under normal conditions**
 - **Incompatible with:** Strong alkalis (forms Zn(OH)₂ precipitate), carbonates, phosphates
 - **Hazardous Decomposition:** Zinc oxide fumes, sulphur oxides (at high temp)
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11. Toxicological Information

- **Acute Toxicity (oral, rat):** LD₅₀ ~2,000–3,000 mg/kg (moderately harmful if ingested in large amounts)
 - **Eye Irritation:** Causes serious eye irritation (due to acidity and zinc ions)
 - **Skin Irritation:** Low; may cause mild irritation with prolonged contact
 - **Inhalation:** Dust may irritate respiratory tract
 - **Carcinogenicity:** Not classified
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12. Ecological Information

- **Aquatic Toxicity:** **Very toxic to aquatic life** (zinc ions are toxic even at low concentrations)
 - **Persistence:** Zinc may accumulate in sediments
 - **Precaution:** Do not discharge into environment. Follow local regulations for agricultural use.
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13. Disposal Considerations

- **Product:** Can be used as intended (fertilizer) in zinc-deficient soils
 - **Waste:** Dispose of in accordance with local hazardous/non-hazardous waste rules (zinc content may trigger regulation)
 - **Packaging:** Rinse thoroughly before disposal or recycling
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14. Transportation Information

- **UN Number:** Not regulated for fertilizer-grade material
 - **Proper Shipping Name:** Zinc sulphate
 - **Transport Classification:** Not restricted under IMDG, IATA, or ADR for agricultural use
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15. Regulatory Information

- **GHS Classification:** H302, H319, H335
 - **Fertilizer (Control) Order, India:** Compliant
 - **REACH:** Registered (if >1 tpa)
 - **TSCA:** Listed
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16. Other Information

This SDS is based on current scientific knowledge and good manufacturing practice. It does not constitute a guarantee. Users must comply with national and local regulations for agricultural chemicals and environmental protection.

Issued by: Quality Control Department, Kesar Petroproducts Ltd.

Date: 29 December 2025